Introduction To Algorithms

Week 8 – Final Synthesis & Assessment:

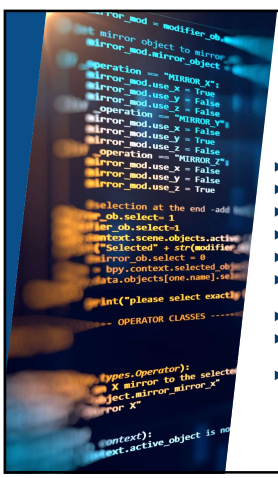
"Explaining the Solution You Submit"

1

Terms to Remember

Term	Brief Meaning
Practical consideration	Real-world issue that affects use
Explainability	Ability to explain how or why a result happened
Global explanation	Explains overall system behavior
Local explanation	Explains one specific output or decision
LIME	Local explainability technique
Bias	Systematic unfair skew or distortion
Discrimination	Harmful unequal treatment
Privacy	Protection of sensitive information
Traceability	Ability to follow evidence and decisions
Inconclusive evidence	Evidence that does not fully prove the claim
Misguided evidence	Evidence that points attention in the wrong direction
Unfair outcome	Result that harms or disadvantages unfairly
CRISP-DM	Data mining process lifecycle
Black swan event	Rare high-impact event hard to predict beforehand

2



What We Will Use Today

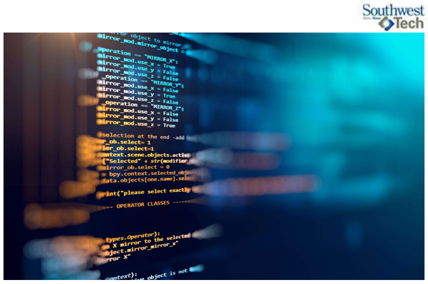
- Explainability
- Evidence quality
- Assumptions and limitations
- Ethical risk recognition
- Bias and privacy cautions
- Cost, time, and accuracy tradeoffs
- Black Swan Event awareness
- FINAL: Part 1 and Part 2 preparation
- Optional bonus-credit / enrichment application

3

What We Will Explore in Other Courses

- Formal AI ethics
- Applied model explainability
- Larger-scale data validation
- Security and privacy governance
- Production monitoring
- Advanced ML model evaluation
- Responsible AI system design

4



Review Coding Lab

5

Review: What Lab 07 Taught Us

Last week - A result depended on choices:

- data representation
- similarity rule
- ranking or grouping method
- assumptions
- limitations
- AI/data connection

6

Today's Question

After an algorithm gives an answer, what still needs to be explained?

7

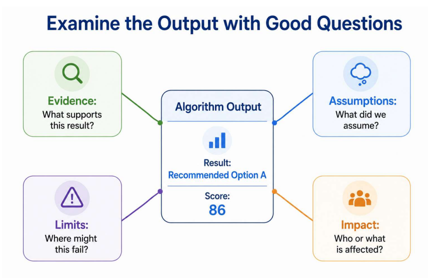


Image Slide

8

Final Week is Synthesis

No Lab!

It is:

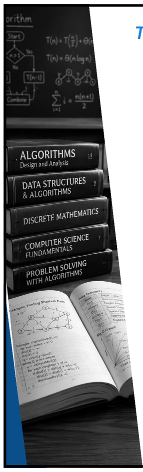
- final synthesis
- practical-risk discussion
- explanation practice
- final assessment preparation & exam

9

What Counts as Success Today

- explain evidence
- name assumptions
- describe tradeoffs
- identify limitations
- discuss responsible AI/tool use
- prepare for the final explanation defense

10



Textbook Review

Practical Considerations

Chapter 16 asks:

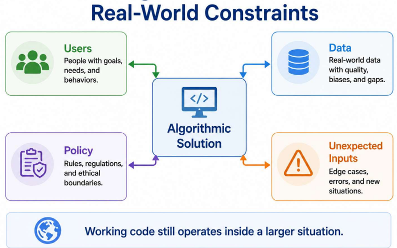
"What happens when algorithms meet real-world use?"

The reading covers:

- - unexpected behavior
- - explainability
- - ethics
- - bias
- - privacy
- - when algorithms fit
- - limits of prediction

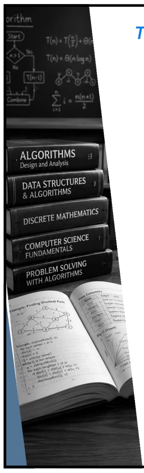
11

Where Algorithmic Code Meets Real-World Constraints



Practical Considerations

12



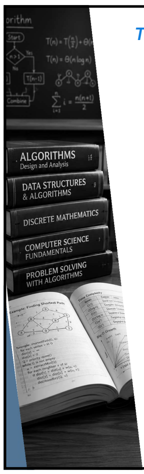
Textbook Review

Expecting The Unexpected

Algorithmic solutions face challenges:

- unusual inputs
- missing data
- changing behavior
- misuse
- unclear goals
- real-world consequences

13



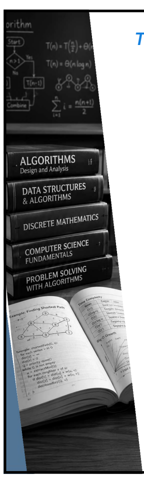
Textbook Review

Cautionary Example: Tay

Tay shows that systems can fail when:

- public input is uncontrolled
- learning behavior is not bounded
- safeguards are weak
- harmful patterns are amplified

14



Textbook Review

Explainability

Explainability asks:

- Why did the system produce this output?
- What evidence supports it?
- What inputs mattered?
- What assumptions shaped it?
- What should we not conclude?

15

Textbook Review

Global And Local Strategies

Global explanation:

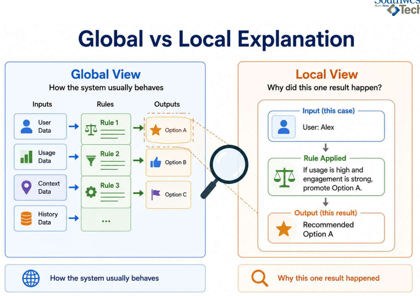
- explains overall behavior

Local explanation:

- explains one specific output or decision

Both can matter.

16



Global and Local Strategies

17

Textbook Review

LIME Recognition

Local Interpretable Model-Agnostic Explanations (LIME)

It is a local explainability technique.

For this course:

- recognize the idea
- do not implement it
- connect it to explaining one result

18

Textbook Review

Ethics And Algorithms

Ethical questions include:

- Bias (& Typicality Bias)
- Discrimination
- Privacy
- Unfair outcomes
- Misleading evidence
- Responsibility for tool use

19

Textbook Review

When Should We Use Algorithms?

Consider:

- cost
- time
- accuracy
- risk
- explainability
- human impact

Not every problem needs an algorithmic answer.

20

Explainability

21

Working Code Is Not The Whole Answer

Working code is valuable evidence.

But we still ask:

- What problem did it solve?
- What cases were tested?
- What did it assume?
- What could fail?
- What tradeoff did it make?

22

Global Explanation

Global explanation describes the overall approach.

Example:

"This solution checks eligibility by evaluating account status, training, equipment availability, and overdue-device status before returning a decision."

23

Local Explanation

Local explanation describes one output.

Example:

"This request returned `needs review` because training was complete, but the device type requires supervisor approval."

24

Evidence Supports Explanation

Useful evidence may include:

- test table
- trace table
- timing table
- comparison table
- ranking table
- README explanation

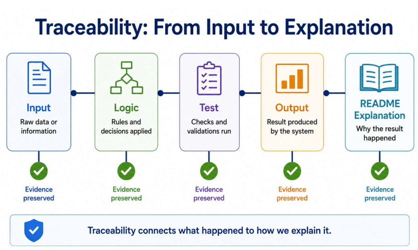
25

Traceability

Traceability means a reviewer can follow:

- input
- logic
- evidence
- output
- explanation

26



Traceability

27

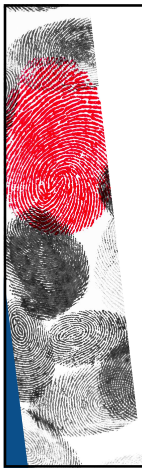
Inconclusive Evidence

Inconclusive evidence does not fully prove the claim.

Examples:

- too few tests
- missing edge cases
- tiny dataset
- only happy-path examples
- unsupported performance claims

28



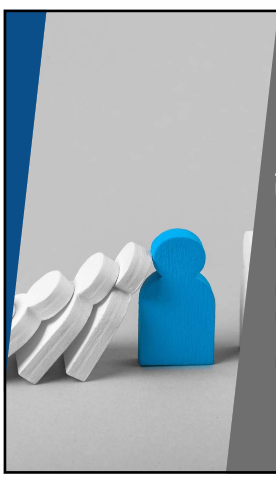
Misguided Evidence

Misguided evidence can look convincing while measuring the wrong thing.

Ask:

- Does this evidence match the claim?
- Is the test realistic?
- Is the dataset representative?
- Is the conclusion too broad?

29



Limitations Are Not Failure

A limitation is an honest boundary.

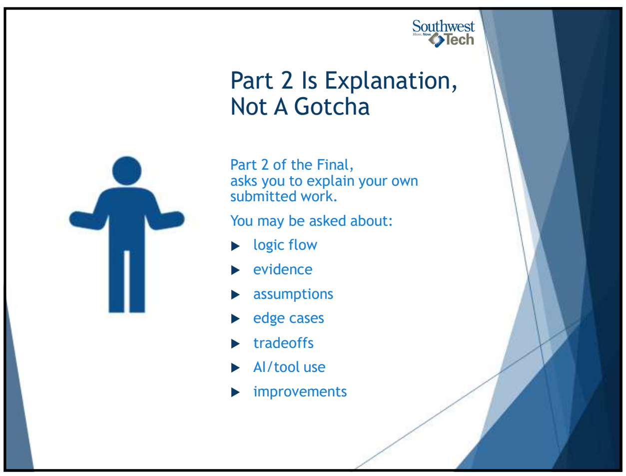
It can explain:

- Where the solution may fail
- What data is missing
- What assumption may not hold
- What should be improved later

30


Tradeoff Explanation
Tradeoffs may involve:
• correctness
• readability
• speed
• memory
• simplicity
• fairness
• explainability

31


Southwest Tech
Part 2 Is Explanation, Not A Gotcha
Part 2 of the Final, asks you to explain your own submitted work.
You may be asked about:
▶ logic flow
▶ evidence
▶ assumptions
▶ edge cases
▶ tradeoffs
▶ AI/tool use
▶ improvements

32


Southwest Tech
Ethics and Evidence Quality

33

Bias And Discrimination

Bias can shape algorithmic results.

Watch for:

- incomplete data
- skewed examples
- unfair criteria
- historical patterns
- hidden assumptions

34

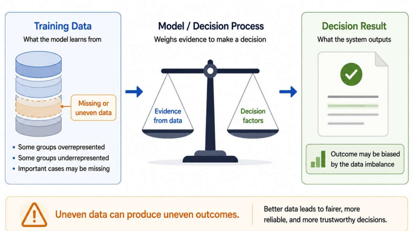


Image Slide

35

Privacy

Privacy asks:

- What data is collected?
- Is it necessary?
- Who can see it?
- How is it protected?
- Could the output reveal sensitive information?

36

Problems With Learning Algorithms

Learning algorithms may:

- learn from poor data
- amplify harmful patterns
- drift over time
- behave differently after deployment
- become difficult to explain

37

Ethical Questions By Context

Ask different questions for:

- classification
- regression
- recommendation
- data mining

Same principle:

What could go wrong, and who is affected?

38

Factors Affecting Algorithmic Solutions

Results can be affected by:

- data quality
- feature choice
- assumptions
- model or algorithm choice
- evaluation method
- deployment context
- human interpretation

39

Unfair Outcomes

An unfair outcome may come from:

- unfair criteria
- missing context
- biased data
- poor thresholds
- over-automation
- weak review process

40

Traceability As Protection

Traceability helps answer:

- What happened?
- Why did it happen?
- What evidence supports it?
- Who reviewed it?
- What should change?

41

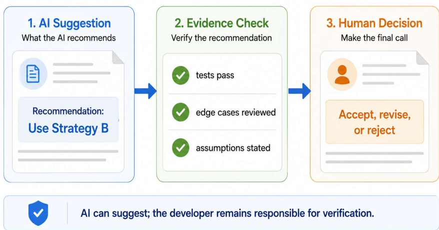
AI Use Accountability

If AI helped, explain:

- What it helped with
- What you changed
- What you tested
- What you rejected
- What you still understand and own

42

Human Review: AI Suggestion vs. Evidence



Human Authority!

43

Previous Work As Reference

You may reference your own previous work.

Be ready to explain:

- What you reused
- What you changed
- Why it fits this task
- How you verified it

44

Reducing Bias Is A Process

Bias reduction is not one switch.

It may involve:

- better problem framing
- better data review
- better evaluation
- more traceability
- human review
- revision over time

45

Break

10 Minute "Human" bio-break

46

Southwest Tech

Practical Use and Unexpected Results

47

Southwest Tech

CRISP-DM Recognition

Cross-Industry Standard Process - Data Mining

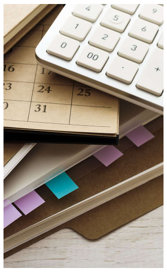
CRISP-DM is a data-mining process *lifecycle*.

Use it as a reminder:

- understand the problem
- understand the data
- prepare the data
- model
- evaluate
- deploy carefully

48

Cost, Time, Accuracy



When deciding whether to use an algorithm, ask:

- What does it cost?
- How much time does it save?
- How accurate must it be?
- What happens when it is wrong?

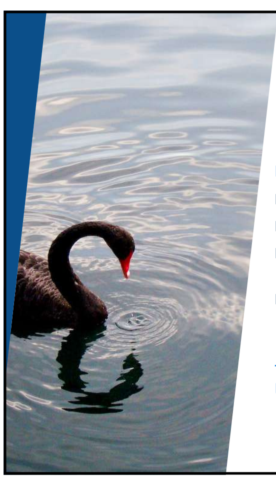
49

Accuracy Is Not Enough

A solution may be accurate but still problematic if it is:

- impossible to explain
- unfair
- too expensive
- too slow
- too brittle
- harmful when wrong

50



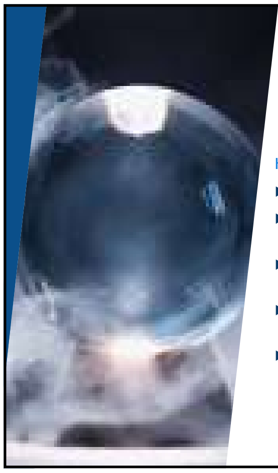
Black Swan Events

Black swan events are:

- unexpected
- high impact
- easier to explain after they happen
- not always surprising to everyone

They expose prediction limits.

51



Forecasting Dilemmas

Hard questions:

- Can this be predicted?
- Can the implications be predicted?
- What data would be needed?
- What uncertainty remains?
- Who must decide anyway?

52

Post-Event Predictability

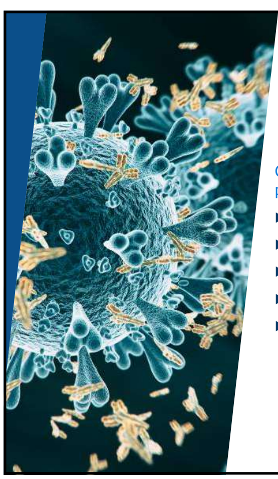
After an event, people may say:

"It was obvious."

But before the event:

- evidence may be incomplete
- signals may be noisy
- implications may be unclear

53

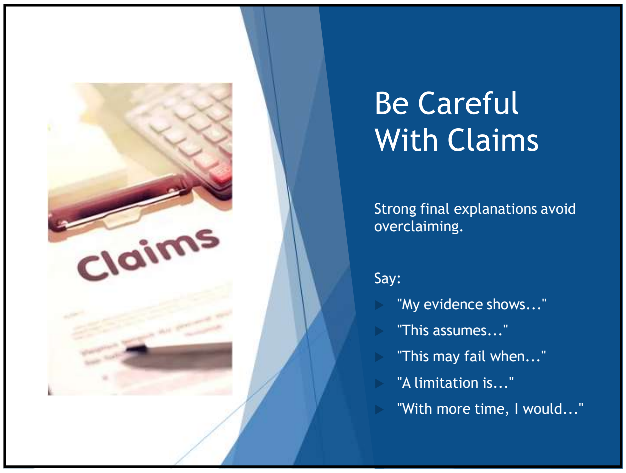


Practical Application: COVID-19

COVID-19 illustrates practical limits:

- changing conditions
- incomplete data
- uncertain implications
- high-stakes decisions
- need for revision over time

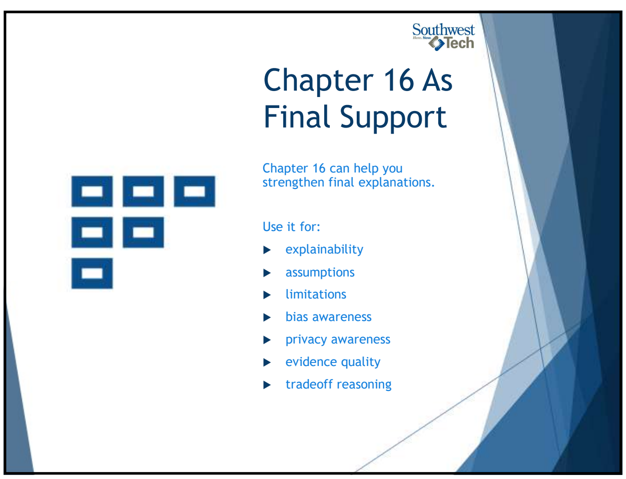
54


Claims
Be Careful
With Claims
Strong final explanations avoid overclaiming.
Say:
"My evidence shows..."
"This assumes..."
"This may fail when..."
"A limitation is..."
"With more time, I would..."

55


Southwest
Tech
Final Bridge &
Optional Bonus

56


Southwest
Tech
Chapter 16 As
Final Support
Chapter 16 can help you strengthen final explanations.
Use it for:
explainability
assumptions
limitations
bias awareness
privacy awareness
evidence quality
tradeoff reasoning

57

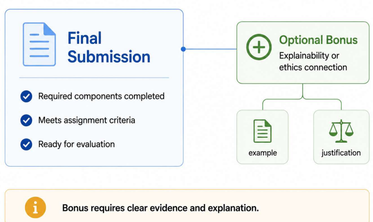
Optional Enrichment / Bonus Credit

You may be able to earn bonus credit by clearly applying a Chapter 16 concept to your final work.
(Instructors Discretion!!)

Examples:
- bias risk
- privacy risk
- inconclusive evidence
- black swan limitation
- cost/time/accuracy tradeoff

58

Optional Bonus Credit as an Evidence-Based Add-On



Optional Enrichment / Bonus Credit

59

Bonus Credit Must Be Demonstrated

Bonus-worthy application should:

- connect to your actual solution
- use evidence
- explain the risk or tradeoff
- avoid vague name-dropping
- appear in README/Part 2 explanation

60

FINAL Exam Structure

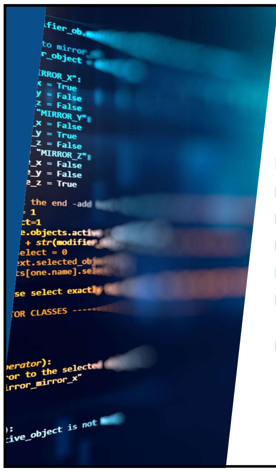
The final has two parts:

- Part 1 - Applied Solution Set
- Part 2 - Explanation Defense

Part 1 is what you submit.

Part 2 is how you explain it

61



Part 1 Readiness

Part 1 should include:

- working solutions
- tests and evidence
- assumptions
- limitations
- comparison where required
- AI/reference-use note when applicable

62

Part 2 Readiness

Be ready to explain:

- what your code does
- why a test passed or failed
- why you chose an approach
- what assumption matters
- what limitation remains
- what AI or previous work contributed

63

Southwest Tech

README As Explanation Prep

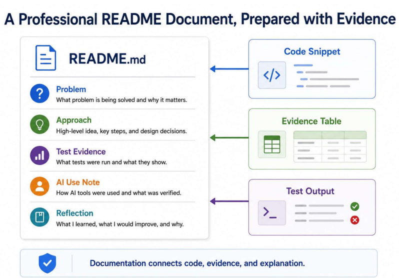
Your README helps you prepare Part 2.

It should organize:

- task summaries
- evidence tables
- assumptions
- limitations
- tradeoffs
- AI/tool-use notes
- previous-work notes

64

Southwest Tech



README Prep

65

Southwest Tech

Final Explanation Pattern

Use this pattern:

1. What did I build?
2. What evidence shows it works?
3. What did I assume?
4. What tradeoff did I make?
5. What limitation remains?
6. What did I verify?

66

Wrap up

67

What To Carry Forward

- algorithms are structured problem solving
- evidence matters
- explanations matter
- assumptions matter
- AI help requires accountability
- working code can be improved

68

Future Reference

After this course, keep using these questions:

- What problem am I solving?
- What data or structure fits?
- How do I know it works?
- What does it assume?
- What could fail?
- How can I explain it clearly?

69

70